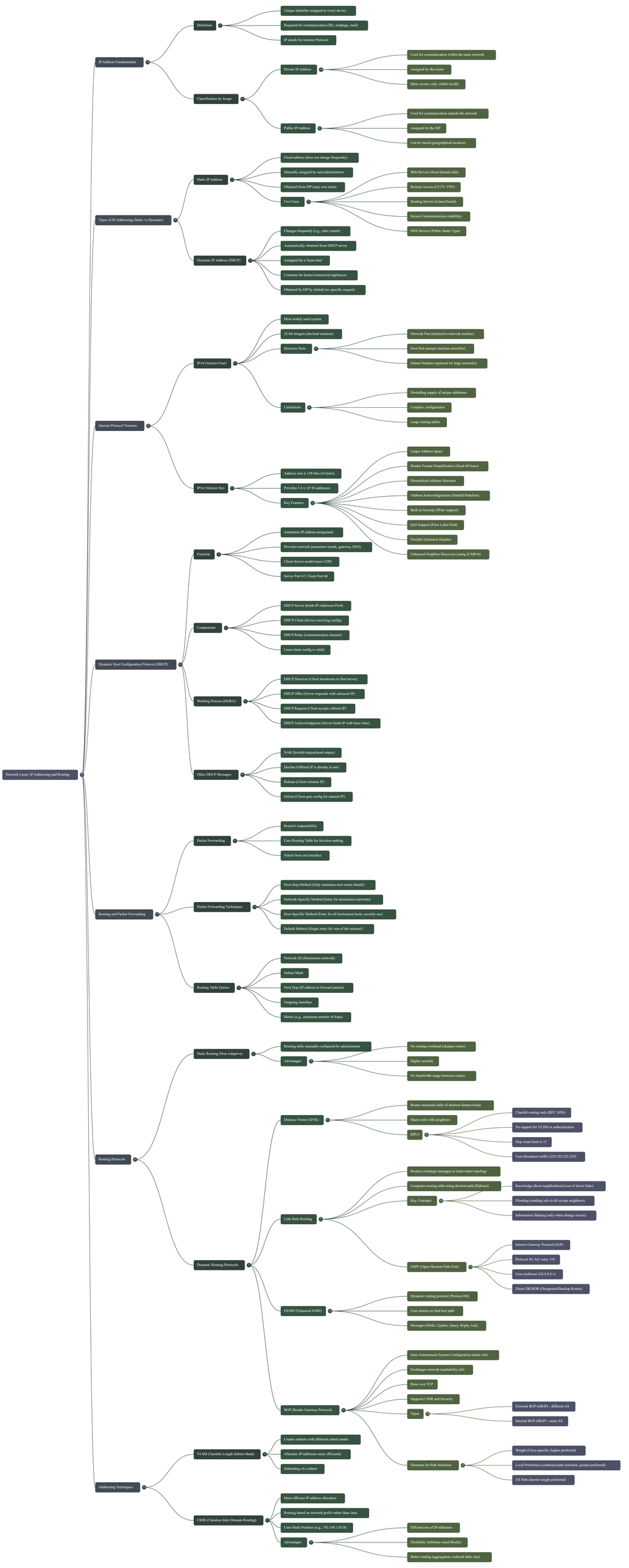
Network Layer: IP Addressing and Routing
IP Address Fundamentals
Definition
Unique identifier assigned to every device
Required for communication (file, webpage, mail)
IP stands for Internet Protocol
Classification by Scope
Private IP Address
Used for communication within the same network
Assigned by the router
More secure; only visible locally
Public IP Address
Used for communication outside the network
Assigned by the ISP
Can be traced (geographical location)
Types of IP Addressing (Static vs Dynamic)
Static IP Address
Fixed address (does not change frequently)
Manually assigned by user/administrator
Obtained from ISP (may cost extra)
Use Cases
Web Servers (fixed domain link)
Remote Access (CCTV, VPNs)
Hosting Servers (Game/Email)
Secure Communications (stability)
DNS Servers (Public Static Type)
Dynamic IP Address (DHCP)
Changes frequently (e.g., after restart)
Automatically obtained from DHCP server
Assigned for a 'lease time'
Common for home/commercial appliances
Obtained by ISP by default (no specific request)
Internet Protocol Versions
IPv4 (Version Four)
Most widely used system
32-bit integers (decimal notation)
Structure Parts
Network Part (distinctive network number)
Host Part (unique machine identifier)
Subnet Number (optional for large networks)
Limitations
Dwindling supply of unique addresses
Complex configuration
Large routing tables
IPv6 (Version Six)
Address size is 128 bits (16 bytes)
Provides 3.4 x 10^38 addresses
Key Features
Larger Address Space
Header Format Simplification (fixed 40 bytes)
Hierarchical Address Structure
Address Autoconfiguration (Stateful/Stateless)
Built-in Security (IPsec support)
QoS Support (Flow Label field)
Flexible Extension Headers
Enhanced Neighbor Discovery (using ICMPv6)
Dynamic Host Configuration Protocol (DHCP)
Function
Automates IP address assignment
Provides network parameters (mask, gateway, DNS)
Client-Server model (uses UDP)
Server Port 67, Client Port 68
Components
DHCP Server (holds IP Addresses Pool)
DHCP Client (device receiving config)
DHCP Relay (communication channel)
Lease (time config is valid)
Working Process (DORA)
DHCP Discover (Client broadcasts to find server)
DHCP Offer (Server responds with unleased IP)
DHCP Request (Client accepts offered IP)
DHCP Acknowledgment (Server binds IP with lease time)
Other DHCP Messages
NAK (Invalid request/pool empty)
Decline (Offered IP is already in use)
Release (Client releases IP)
Inform (Client gets config for manual IP)
Routing and Packet Forwarding
Packet Forwarding
Router's responsibility
Uses Routing Table for decision making
Selects best exit interface
Packet Forwarding Techniques
Next-Hop Method (Only maintains next router details)
Network-Specific Method (Entry for destination networks)
Host-Specific Method (Entry for all destination hosts, security use)
Default Method (Single entry for 'rest of the internet')
Routing Table Entries
Network ID (Destination network)
Subnet Mask
Next Hop (IP address to forward packet)
Outgoing Interface
Metric (e.g., minimum number of hops)
Routing Protocols
Static Routing (Non-Adaptive)
Routing table manually configured by administrator
Advantages
No routing overhead (cheaper router)
Higher security
No bandwidth usage between routers
Dynamic Routing Protocols
Distance Vector (DVR)
Router maintains table of shortest distance/hops
Shares info with neighbors
RIPv1
Classful routing only (RFC 1058)
No support for VLSM or authentication
Hop count limit is 15
Uses Broadcast traffic (255.255.255.255)
Link State Routing
Routers exchange messages to learn entire topology
Computes routing table using shortest path (Dijkstra)
Key Concepts
Knowledge about neighborhood (cost of direct links)
Flooding (sending info to all except neighbors)
Information Sharing (only when change occurs)
OSPF (Open Shortest Path First)
Interior Gateway Protocol (IGP)
Protocol #9; AD value: 110
Uses multicast 224.0.0.5/.6
Elects DR/BDR (Designated/Backup Router)
EIGRP (Enhanced IGRP)
Dynamic routing protocol (Protocol 88)
Uses metrics to find best path
Messages (Hello, Update, Query, Reply, Ack)
BGP (Border Gateway Protocol)
Inter-Autonomous System Configuration (main role)
Exchanges network reachability info
Runs over TCP
Supports CIDR and Security
Types
External BGP (eBGP) - different AS
Internal BGP (iBGP) - same AS
Elements for Path Selection
Weight (Cisco specific, higher preferred)
Local Preference (outbound path selection, greater preferred)
AS Path (shorter length preferred)
Addressing Techniques
VLSM (Variable Length Subnet Mask)
Creates subnets with different subnet masks
Allocates IP addresses more efficiently
Subnetting of a subnet
CIDR (Classless Inter-Domain Routing)
More efficient IP address allocation
Routing based on network prefix rather than class
Uses Slash Notation (e.g., 192.168.1.0/24)
Advantages
Efficient use of IP addresses
Flexibility (arbitrary-sized blocks)
Better routing (aggregation, reduced table size)